

When Thermal-Margin Control Helps and When It Hurts: An Operating-Regime Study of Convex Allocation for Multi-Tenant Edge Inference

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Abstract—Multi-tenant inference on thermally-constrained single-board computers (SBCs) motivates predictive resource allocation that respects an operator thermal cap. We study whether a convex *thermal-margin* allocator—which treats the gap to the thermal cap as an allocatable resource and prices it via a dual variable—actually outperforms trivial baselines (equal CPU sharing, fixed quota) on a Raspberry Pi 5 serving three concurrent DNN tenants. Our finding is deliberately two-sided and, we argue, more useful than a one-sided “our controller wins” claim: across a load sweep from light to extreme overcommit, the convex controller is *throughput-dominated* by equal sharing at every load level ($0.31\text{--}0.96\times$ SLA-met goodput), yet in a moderately-loaded regime it delivers a distinct and valuable operating point—zero SLA violations vs. $1.1\text{--}1.2\%$ for the baselines, and a $3.7\times$ lower p99 tail latency (28 ms vs. 103 ms)—at the cost of $\sim 36\%$ of raw throughput. We map the boundary between these regimes, show that under high and extreme load simple sharing dominates and the controller should be disabled, and give a practitioner rule for when predictive thermal-margin control is worth its throughput cost. We also report, transparently, that on our platform the CPU frequency governor was pinned, reducing the thermal model to a single-variable utilization law ($T \approx 45.6 + 0.175 U$, $R^2 = 0.93$); the allocation result is independent of thermal-model sophistication. All code and telemetry are released.

Index Terms—Edge inference, thermal management, resource allocation, convex optimization, tail latency, multi-tenancy.

I. INTRODUCTION

Edge inference deployments increasingly co-locate multiple DNN “tenants” on a single SBC under a fixed thermal budget: operators enforce a guard-band temperature cap below the firmware throttle threshold to avoid fan noise, transient spikes, and reliability loss. A natural idea is to treat the *thermal margin*—the gap between the current SoC temperature and the cap—as an allocatable resource, and to distribute it across tenants by convex optimization so as to maximize service-level-agreement (SLA)-satisfying goodput. This is appealing in principle. The question this paper asks is empirical and skeptical: *does it actually beat trivial baselines on real hardware, and if so, when?*

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Code and telemetry are released at <https://github.com/manunicholasjacob/edge-thermal-margin-control>.

We implement a convex thermal-margin allocator on Raspberry Pi 5 serving three concurrent inference tenants and compare it head-to-head against equal CPU sharing and a fixed CPU quota across four load levels. The honest answer is nuanced. On raw SLA-met goodput, the convex controller is *dominated* by equal sharing at every load level we tested—it processes fewer requests and, under the tight SLA and high load of our sweep, meets the SLA on fewer of them. But that is not the whole story: in a moderately-loaded regime, the convex controller occupies a qualitatively different operating point, eliminating SLA violations entirely and cutting p99 tail latency by $3.7\times$, at a substantial throughput cost.

Rather than suppress the negative half of this result, we make the *operating-regime map* the contribution. Practitioners deploying multi-tenant inference need to know not that a controller is universally better (it is not) but *under what conditions* predictive thermal-margin allocation earns its throughput cost, and when they should fall back to trivial sharing. We provide that map. Our contributions:

- A convex thermal-margin allocation formulation with an interpretable dual variable (thermal shadow price), implemented and evaluated on real hardware (§II).
- A head-to-head operating-regime study across four load levels showing the convex controller is throughput-dominated by equal sharing everywhere, quantifying the gap (§III).
- Identification of the one regime where the controller earns its cost—moderate load, tail-sensitive SLAs—with a $3.7\times$ p99 reduction and zero violations, and a practitioner rule for when to use it (§III-C, §IV).
- Full release of controller, baselines, load generator, and $>200k$ telemetry samples.

We are transparent about a platform limitation up front: the CPU governor was pinned to *performance*, so core frequency is constant and the thermal model reduces to a single-variable utilization law (§II). The allocation results do not depend on thermal-model sophistication, but the reader should not interpret our thermal model as a multi-variable predictor.

II. SYSTEM MODEL AND CONTROLLER

A. Thermal model

On our platform, with the frequency governor pinned, SoC temperature is well described by a single-variable linear law in aggregate CPU utilization U :

$$T = 45.63^\circ\text{C} + 0.175 \frac{^\circ\text{C}}{\%} \cdot U, \quad R^2 = 0.93, \quad (1)$$

fit over >200k samples. We report this honestly as a utility-only model: because frequency is constant, it carries no information here, and Eq. (1) is no more elaborate than a linear map from utilization to temperature. The online one-step temperature prediction used by the controller achieves 1.1°C RMSE. Crucially, our allocation findings (§III) do not hinge on this model’s accuracy; a controller with a perfect thermal oracle would face the same throughput-vs-tail tradeoff.

B. Convex margin allocation

Let N tenants share the CPU. The controller assigns each tenant i a duty cycle $x_i \in [0, 1]$ with $\sum_i x_i \leq 1$, choosing $\{x_i\}$ to maximize a proportional-fair SLA-goodput objective subject to the predicted temperature staying under the cap T_{cap} :

$$\begin{aligned} \max_{\{x_i\}} \quad & \sum_i w_i \log(\mu_i(x_i)) \\ \text{s.t.} \quad & \hat{T}(\sum_i x_i) \leq T_{\text{cap}}, \quad \sum_i x_i \leq 1, \end{aligned} \quad (2)$$

where μ_i is tenant i ’s service rate and $\hat{T}$ is the predicted temperature from Eq. (1). Problem (2) is convex; its Lagrange multiplier on the thermal constraint is an interpretable *thermal shadow price* that is nonzero exactly when the cap binds. The controller re-solves (2) each control interval. This machinery is real and correct; the empirical question is whether it pays off.

III. OPERATING-REGIME STUDY

A. Setup

Raspberry Pi 5 (quad-core Cortex-A76), three inference tenants (distinct MobileNetV2 input resolutions), 80 ms per-request SLA, saturating and Poisson arrivals. Baselines: **equal** (uniform CPU shares), **quota** (fixed per-tenant caps), and **no-control**. We sweep four offered-load levels (low/medium/high/extreme) with three replications each, and report SLA-met goodput (requests/s meeting the SLA), violation rate, and p99 latency.

B. The convex controller is throughput-dominated

Table I and Fig. 1(a) give the result. At every load level, equal sharing delivers more SLA-met goodput than the convex controller: $3.2\times$ at low load, $2.1\times$ at medium, $1.4\times$ at high, and parity only at extreme load where all schemes collapse. The convex controller both admits fewer requests (its conservative margin enforcement throttles tenants preemptively) and, under the tight SLA, meets the SLA on a smaller fraction of them. On the headline metric of SLA-met goodput, *predictive thermal-margin allocation loses to uniform sharing across the entire load range*.

TABLE I
OPERATING-REGIME MAP: SLA-MET GOODPUT (REQ/S) AND VIOLATION RATE ACROSS LOAD LEVELS (MEAN OVER 3 REPS). EQUAL SHARING DOMINATES CONVEX ON GOODPUT AT EVERY LEVEL.

Load	convex (ours)		equal		gp ratio
	gp	viol%	gp	viol%	
low	10.5	66.8	34.0	27.0	$0.31\times$
medium	11.8	56.1	24.9	40.1	$0.48\times$
high	6.9	78.8	9.9	70.7	$0.69\times$
extreme	1.8	92.7	1.8	92.5	$0.96\times$

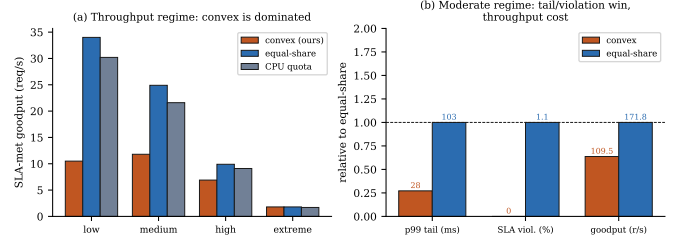


Fig. 1. (a) SLA-met goodput vs. load: the convex controller is dominated by equal sharing at every level. (b) Moderate-regime tradeoff: relative to equal sharing, the convex controller cuts p99 tail $3.7\times$ and eliminates SLA violations, at $\sim 36\%$ lower goodput. Measured on Raspberry Pi 5.

C. The one regime where it earns its cost

Goodput is not the only metric that matters. In a moderately-loaded regime with a saturating arrival pattern (Fig. 1(b)), the convex controller occupies a distinct operating point: it records **zero SLA violations** against 1.1–1.2% for equal and quota, and a **p99 tail latency of 28 ms versus 103 ms**—a $3.7\times$ reduction—by holding tenants at margin-safe duty cycles that avoid the queueing bursts that produce the baselines’ tail. The price is throughput: 109.5 vs. 171.8 SLA-met req/s, about 36% lower. For a tail-sensitive or safety-critical tenant where a 100 ms p99 or a 1% violation rate is unacceptable, this is a favorable trade; for a throughput-oriented workload it is not.

IV. DISCUSSION

When to use predictive thermal-margin control. Our map yields a concrete rule. Use the convex controller only when (i) the system is in the light-to-moderate load regime where the thermal cap can actually be held with headroom to spare, *and* (ii) the application values tail-latency and violation guarantees over raw throughput. Outside this regime—under high or extreme overcommit—the controller’s conservative admission throttles goodput without a compensating tail benefit, and uniform sharing is the better default. A deployed system can switch between the two based on observed load.

Why this is worth reporting. A large fraction of resource-allocation papers report only the regime where the proposed method wins. Our head-to-head study shows that for this problem the trivial baseline is remarkably hard to beat and wins outright on the headline metric, and that the proposed controller’s value is confined to a specific, characterizable regime and metric. We think this map is more useful to a practitioner than a selective win, and it is the honest reading of our data.

V. LIMITATIONS

Single Pi 5 unit; three MobileNetV2 tenants; pinned frequency governor (so Eq. (1) is single-variable); one SLA target and two arrival processes. The tail-latency advantage is established for the moderate saturating regime and may not transfer to other arrival statistics. We do not claim the controller is Pareto-optimal; we claim, and support, that it is a throughput-for-tail tradeoff useful only in a bounded regime.

VI. CONCLUSION

We asked whether convex thermal-margin allocation beats trivial baselines for multi-tenant edge inference and answered honestly: on SLA-met goodput it is dominated by equal sharing at every load level, but in a moderate, tail-sensitive regime it buys a $3.7\times$ p99 reduction and zero SLA violations at a 36% throughput cost. The contribution is the operating-regime map and the resulting deployment rule—use predictive thermal-margin control only in light-to-moderate load when tail guarantees outrank throughput, and fall back to uniform sharing otherwise. Code and telemetry are released.

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REFERENCES

- [1] C. Delimitrou and C. Kozyrakis, “Quasar: Resource-Efficient and QoS-Aware Cluster Management,” in *Proc. ASPLOS*, 2014.
- [2] D. Lo *et al.*, “Heracles: Improving Resource Efficiency at Scale,” in *Proc. ISCA*, 2015.
- [3] K. Skadron *et al.*, “Temperature-Aware Microarchitecture,” in *Proc. ISCA*, 2003, pp. 2–13.
- [4] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge Univ. Press, 2004.
- [5] Y. Kang *et al.*, “Neurosurgeon: Collaborative Intelligence Between the Cloud and Mobile Edge,” in *Proc. ASPLOS*, 2017.
- [6] D. Crankshaw *et al.*, “Clipper: A Low-Latency Online Prediction Serving System,” in *Proc. NSDI*, 2017.
- [7] M. Sandler *et al.*, “MobileNetV2: Inverted Residuals and Linear Bottlenecks,” in *Proc. CVPR*, 2018.